

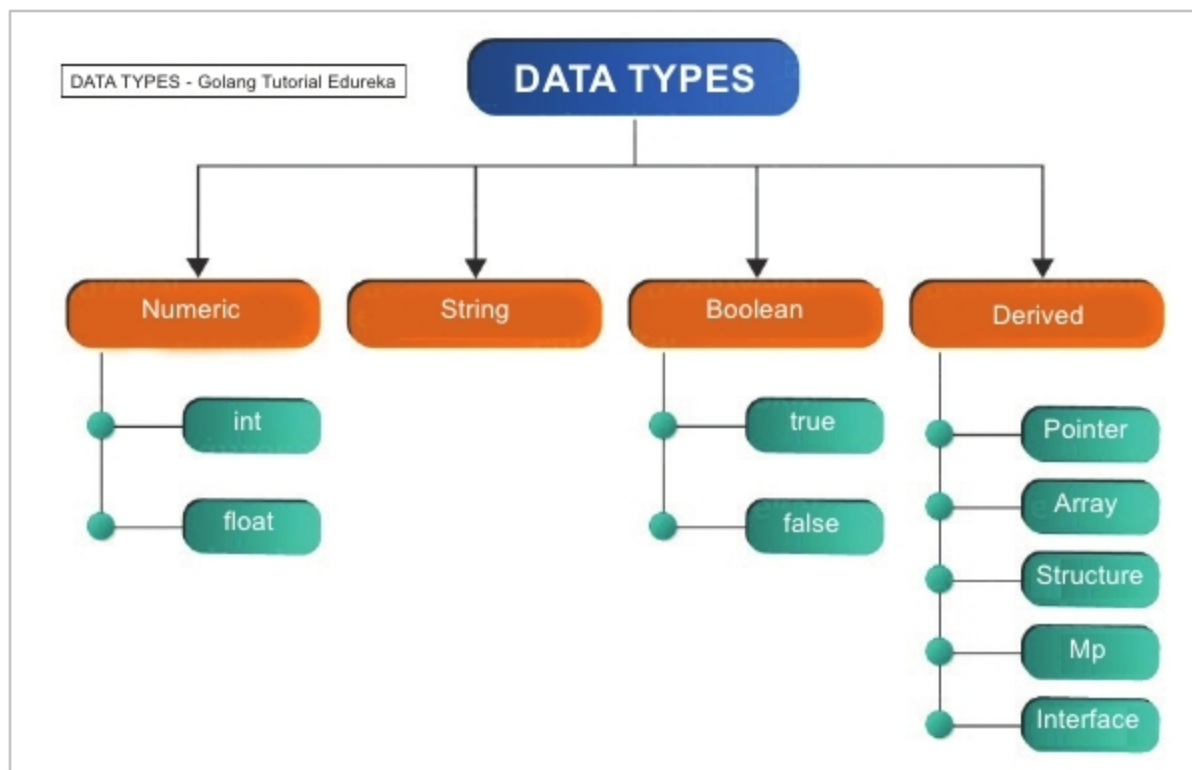


Figure 1: Data types

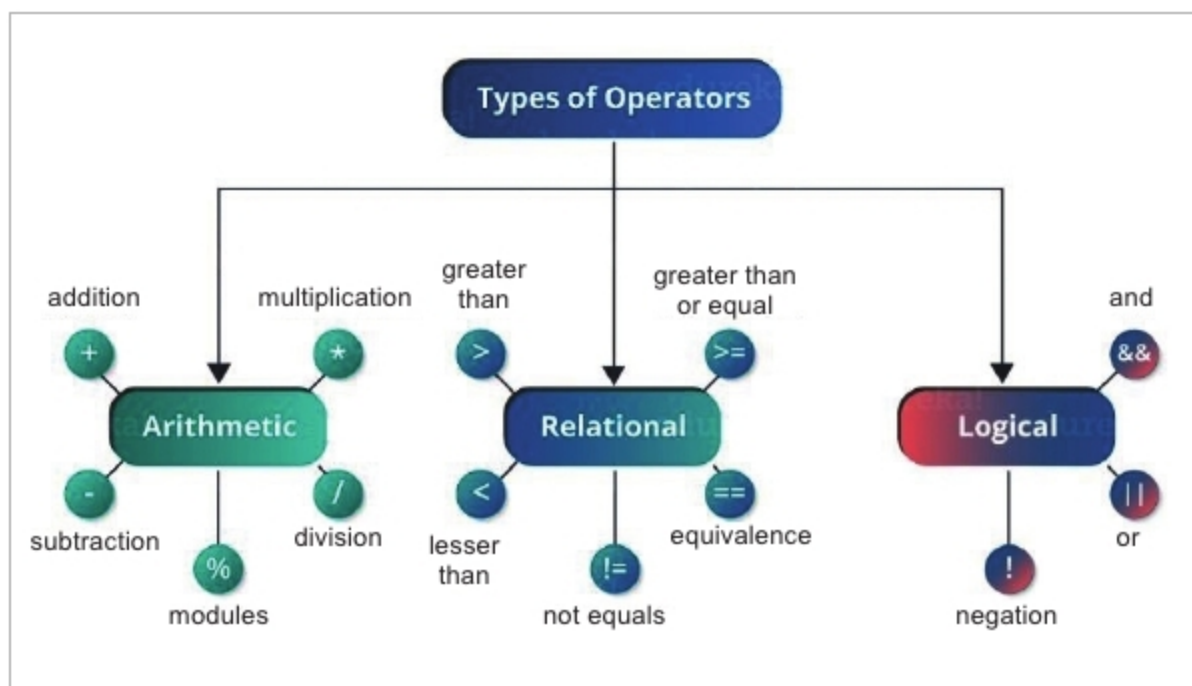


Figure 2: Operators

```

// used for comments
fmt.Println("Hello, World!")
}

```

OUTPUT: Hello,World!

Let's examine the above Go program line by line.

Line 1: This contains the *package main* of the program, which has the overall content of the program. It is the opening point to run the program; so it is compulsory to write this.

Line 2: This contains *import fmt*, which is a preprocessor command that tells the compiler to incorporate the files

lying in the package.

Line 3: This is the *main* function — the beginning of the execution of the program.

Line 5: *fmt.Println()* is a standard library function to print something as an output on the screen. In this case, the *fmt* package has transmitted the *Println* method, which is used to show the output.

Note: Comments are used for explaining code and are used in Java, C or C++. Compilers ignore the comment entries and do not execute them. Comments can be of a single line or multiple lines.

Go, Java and C++: A comparison

Here is a comparison of Go with Java and C++.

- Go is a procedural and concurrent programming language, while Java and C++ are object-oriented programming languages.
- Go does not contain classes with constructors and destructors, while both Java and C++ contain classes with constructors and destructors.
- Go contains pointers, but does not contain an arithmetic pointer. Java doesn't support pointers, while C++ contains both pointers as well as arithmetic pointers.
- In Go, maps are passed by reference, while in Java and in C++ maps are passed as a value.
- Go does not support exception handling; instead it has errors. Both Java and C++ support the concept of exception handling.

Data types and operators

Data types: The data types supported by Go are shown in Figure 1.

Operators: Go has three general types of operators that are prevalent in every major programming language, as shown in Figure 2.

New features in the latest stable version

The latest stable version of Go is 1.13.1, which was released on September 3, 2019. Its new features are mentioned below.

- Go now supports a more uniform and modernised set of number literal prefixes.
- Support for TLS 1.3 in the *crypto/tls* package is available by default (opt-out will be removed in Go 1.14).
- It now has support for error wrapping. **END** 🐧

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